

Mobile sink assisted data gathering for URLLC in IoT using a fuzzy logic system

Seyed Salar Sefati

University polytechnic of Bucharest
Electronic, Telecommunications and Information Technology
Bucharest, Romania
Sefati@ieee.org

Simona Halunga

University polytechnic of Bucharest
Electronic, Telecommunications and Information Technology
Bucharest, Romania
Simona.halunga@upb.ro

Abstract—The Internet of Things (IoT) is a new technology that employs a variety of sensors and wireless communication protocols. People are leveraging IoT facilitate their actions by using innovative and intelligent equipment. An increasing number of sensors can decrease the Quality of Service (QoS), so implementing a suitable algorithm in an IoT network may enhance this parameter. One of the important applications in IoT with strict QoS requirements is the Ultra-Reliability and Low Latency Communication (URLLC). Several types of URLLC services can be provided in the IoT based on the user's requirements. Also, a proper Cluster Head (CH) selection plays a vital role in receiving and transmitting data and reduces the energy consumption of the devices. In today's IoT devices, the mobility of the sink nodes is one of the most important features to be considered in real-world applications. The selection of the CH is an NP-Hard problem; thus, a fuzzy logic protocol is used to select the appropriate CH and intermediate nodes for routing, and the results are presented in this paper. This method is developed in order to minimize the delay, and energy consumption as well as enhancing the reliability. The method has been tested via MATLAB Simulation and the results are presented in the final section of the present paper.

Keywords— Internet of Things, fuzzy logic, Ultra reliability and Low latency (URLLC), Mobile sink

I. INTRODUCTION

Around 35 billion intelligent devices, including cellphones, medical, transportation, Industry, and utility organizations, are linked to the Internet, and this number is rapidly expanding. Many devices and different types of sensors can be now interconnected through the Internet, allowing the subscribers to have control over them remotely, creating the so-called Internet of Things (IoT) [1]. The goal of the IoT is to allow connection everywhere, at any time, and with every device connected to the internet [2].

Many IoT scenarios, such as automation in the industrial Vehicle-to-Anything (V2X) networks, grid computing, and remote surgery, may demand an Ultra-Reliable and Low Latency Communication facilities (URLLC) [3]. Two approaches are used to develop URLLC applications in the IoT networks: one is made based on a technological standard for

using shared infrastructure to increase the URLLC performance of public mobile cellular networks in order to satisfy the needs of various IoT applications. The second method is to develop options in existing systems or even dedicated networks for allowing the essential applications and, then, extensions for non-essential ones [4].

The mobile sink plays a vital role in IoT applications, the Cluster Head (CH) gathers the data from other sensors and the mobile sink sends a request to the CH to share its data in the network [5]. In this work we assume that the mobile sink has an unlimited battery and superior computing capacities. Separate mobile sinks may originate different requests [6], but, at this point, we are concerned with optimizing the connection between one mobile sink and the CH. The mobile sink can decrease the communication overhead and reduce energy usage [7]. On the other hand, the network will perform better when the sink is a mobile one, so, by optimizing the sink path we can increase the network lifespan.

This article presents and analyzes the performances of a novel CH selection method based on fuzzy logic. All the network nodes are considered smart elements in charge of sending data to the CH. In order to select the most appropriate CH, we took into consideration several parameters as follows.

The centrality of the CH position within the cluster: when performing the clustering, if the CH is not in the middle of the group of sensors, the normal sensor will encounter difficulties in transferring the data to the CH.

The distance between the CH and mobile sink: If the CH distance between the mobile sink is large, the CH can have difficulties in transferring the data and will waste significant energy for that task.

The number of rounds in which a node is selected as CH: One of the critical issues in IoT is load balancing and energy consumption. CH usually wastes a lot of energy because it has to gather the data from all the other sensors and has to send it to the mobile sink, leading to a high energy consumption.

The energy level at the nodes: The energy consumed by each node is important in an IoT network because many sensors have limited capacity and cannot be recharged.

The use of this method may bring many favorable effects in comparison to other approaches, reflected by

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- a reduced latency and energy consumption, thus consequently enhancing reliability;
- the possibility of selecting the optimal CH with a higher reliability and lower delay in comparison with other algorithms;
- the possibility to accomplish the best placement for IoT in the cluster areas based on fuzzy algorithms.

The remainder of the article is organized as follows: Section II examine, describes, and expands the suggested method; Section III presents and analyzes the experimental simulation settings and findings. Finally, Section IV closes the study with several conclusions and provides recommendations for future research.

II. LITERATURE REVIEW

The idea of employing a mobile sink in IoT has been discussed in the literature. Various researchers have used the mobile sink in the IoT networks, where different parameters of the nodes, evaluated through measurements, are assembled and sent to the mobile sink node.

Yu, Zhang [8] describe ring routing protocol, based on a mobile sink system for WSNs with a random pattern. In their proposed algorithm, a number of sensors are chosen as CH, which store and communicate information regarding their location as well as data through the mobile sink. This paper increases the network lifetime but suffers from low reliability and low latency.

Wang and Chen [9] proposed an effective path designing for consistent data collecting algorithm, named EARTH, that enables the possibility of establishing routes with low latency. The proposed method used the location for mobile sink and the data gathering in CH with suitable location to reduce computational effort and packet loss. However, in this paper the authors assumed that the sensors have many sensing rates but a fixed buffer size. The CHs are in charge of obtaining data from various sensors, which may help to solve the hotspot issue. This paper decreases the energy consumption and low buffer queue but suffers from low availability and reliability.

Tabibi and Ghaffari [10] have suggested a tree-based algorithm for clustering the network and selecting the best location for mobile sink node. The proposed method used a tree-based structure clustering algorithm to find the optimal mobile sink in order to construct high reliability and high throughput among all nodes. but suffers from low load balancing and high energy consumption.

Cheng, Zhang [11] proposed an adaptable control algorithm for merging numerous data flows and fast calculation of new routes to avoid blockings. In their approach, each disordered traffic flow is reassigned to a rerouted path for congestion avoidance. As a result, all data traffic flows that initially had been allocated to the same paths are combined into a single large data flow, reducing thus the amount of changed redirecting rules that are required during the restoration process. This paper increases the packet delivery rate, but suffers from high energy consumption.

Sadrishojaei, Navimipour [12] have suggested a CH selection based on mobile internet of things (MIoT) in sizable

scalable and crowded area. In this paper the CHs are selected according to energy reduction, covering range and best services. The authors applied the random waypoint mobility (RWM) model to replicate a Mobile IoT-based surveillance structure. This proposed method has a good performance in low latency and low energy consumption, but suffers from availability.

Wu, Chen [13] proposed a method that allows forwarding the data from the source nodes to the destination ones by optimizing the average transfer duration and the stability of the queues. Distributed speed control and routing algorithms were used in this suggested technique to lower the mobile sink processing burden and produce a near-optimal data gathering solution. Their research reveals a link between the network efficiency and the average queuing size of the data. This proposed method has a good performance in energy consumption and low buffer queue size, but suffers from reliability and availability.

III. PROBLEM STATEMENT

In terms of the set of QoS parameters in general purpose IoT networks, URLLC may not be at the top of the priority list [14]. URLLC requires that the data transferred from Exchange to Exchange (E2E) is highly reliable, low latency, and highly secure. As a result of these criteria, the 3rd Generation Partnership Project (3GPP) established an ensemble of QoS constraints for URLLC, such as a maximum 1 ms delay and a minimum 99.999% system reliability. The presented method selects efficiently the CH in order to transfer the data to the mobile sink, achieving low latency and high reliability performances by using the fuzzy logic algorithm. In the beginning, the mobile sink is positioned in the middle of the IoT network. Then it transmits a single message to CH that contains the location of the sink, and the speed of the sink. After CH receive this message, all the nodes have to return with a reply message, that contains their IDs and positions. The procedure is depicted in figure 1. The mobile sink moves to the nearest CH to collect the information, and then, finds the closest other CH and starts moving towards it.

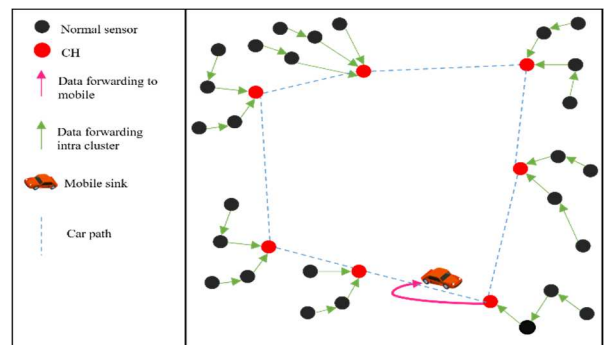


Fig. 1. IoT network using the mobile sink

A. Fuzzy logic algorithm

In everyday life, we sometimes encounter situations where we do not know the right or wrong decision, and the right action not an obvious one. At this point, fuzzy logic offers a flexible and valuable offer. A fuzzy logic system evaluates a collection of input data and generates output data based on an ensemble of

previously established rules. The implementation of such fuzzy logic system involves four major sections: the fuzzifier block, the set of rules, the inference engine and the de-fuzzifier block, as shown in figure 2. A set of input data is at first received by the structure, and the fuzzifier bloc transforms these data into a fuzzy class using the fuzzy base variables or linguistics and membership functions. After that, the inference operation is performed, based on a set of rules. Finally, the de-fuzzifier implements the conversion from the fuzzy set to a single value for each output of the inference block. The fuzzifier converts a real-time crisp input into a fuzzy function, assessing the input's degree of membership' in a nebulous idea. In other words, fuzzy inference creates the correspondence between a given input value to a set or domain by means of fuzzy reasoning. Defuzzification is the reverse process that converts a quantifiable result in sharp logic values based on the preestablished fuzzy sets and membership domains.

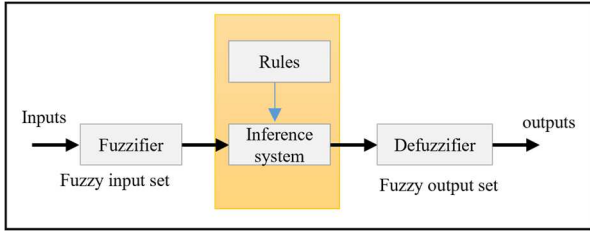


Fig. 2. Fuzzy logic design

B. The parameters of QoS

The QoS express those characteristics of services that are closely related to a certain user and its particular applications like reliability, latency, energy utilization etc. In most applications the latency, and energy consumption have to be reduced, while the reliability has to be increased. The computational effort is closely related to the power consumption.

The time delay evaluated at the physical layer, denoted with τ_1 , is given by (1), and is the sum between the time to transmit delay (τ_t) the transmission time necessary to communicate the data from source to destination τ_{prop} the encoding and decoding processing time τ_{proc} the mean retransmission time necessary when the data has been erroneously received τ_{ret} and the receiver preprocessing time τ_{sig} [15]:

$$\tau_1 = \tau_t + \tau_{prop} + \tau_{proc} + \tau_{ret} + \tau_{sig} \quad (1)$$

For an URLLC system, this physical layer latency should not exceed 0.5 ms.

One of the most basic functions used for modelling the evolution in time domain of the electrical devices is the reliability function, defined by :

$$R_c(t) = e^{-\alpha t} \quad (2)$$

where α is a measure of failure that can be evaluated for each component per time unit, T, evaluated in hours.

The power consumption of the proposed model is computed based on the model, which was introduced in Low-energy adaptive clustering hierarchy (LEACH). The LEACH algorithm is a famous routing protocol in IoT and Wireless Sensor Network (WSN). The energy consumption has a significant impact on the devices that contain IoT potential services. Following that, each possible service must provide a parameter to indicate how much energy it uses [16]:

$$E = \frac{E_{init} - E_c}{E_{init}} \quad (3)$$

where E_{init} , shows the initial energy of each node, and E_c represents the energy of each node at the evaluation time instant.

C. Proposed method

Clustering is a data collecting approach used in IoT that minimizes energy usage by grouping nodes. All the connections with the elements inside or outside the cluster are established via the CH, which controls all Cluster Member (CM) nodes. The evaluation of CH selection is assessed in the inference stage using fuzzy rules based on a set of considerations, such as the fact that the CH is centralized, the distance between the CH and the mobile sink, the number of rounds in which the CH works as a master, and the energy level of nodes' battery is minimized. Four fuzzy sets were defined for each of the four input parameters, resulting in 16 fuzzy rules that are depicted in figure 3.

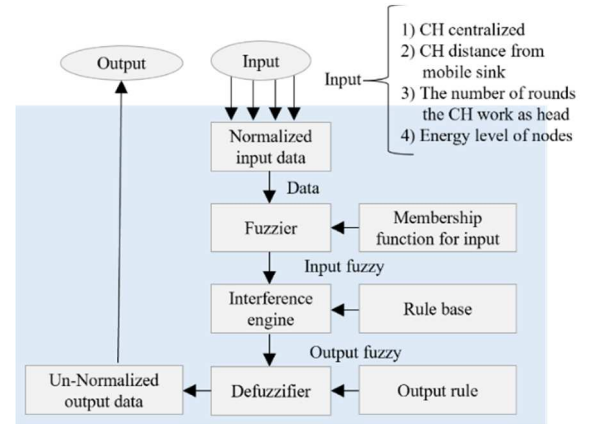


Fig. 3. The block diagram of fuzzy system used in the proposed method

One of the parameters for selecting the appropriate CH is the residual energy of a sensor node. Another essential feature is the centrality of the CH. We assumed at the beginning that, according to figure 1, all the CHs are in the central part of cluster and close to the route of the mobile sink. If the CH in a cluster is in its best location, centralized with respect to the other sensor nodes, the data may be forwarded to CH using less energy. Another important criterion is the distance between the CH and mobile sink; if the mobile sink is moving far from the CHs, they have to spend a large amount of energy to transfer the data towards the mobile sink. The last feature is the number of times a node is selected as a CH. If one node is selected

repeatedly as a CH, the energy of CH is depleted, so the CH needs load balancing. The fuzzy output was converted to a numerical number via defuzzification. Defuzzification was employed in the suggested technique, which was evaluated using formula [17],

$$F(x, w) = \sum_{i=1}^m (x_i * w_i) + \sum_{j=\frac{m}{2}+1}^m ((1-x_i) * w_j) \quad (4)$$

where $x = (x_1, x_2, x_3, x_4)$ represents the set of parameters chosen to select the CH i.e., centrality, distance to the mobile sink, residual energy, and the number of rounds a certain node is selected as the head, respectively. $w = (w_1, w_2, w_3, w_4)$ are the selected weights for the x properties, and m is the length of the weight vector. Defuzzification has been used to transform the data generated by the fuzzy inference engine into a numerical value. In the proposed method, fuzzy output was used which was measured based on [18]:

$$Fuzzy \text{ output} = \frac{\sum_{i=1}^m y^{-1} \prod_{i=1}^n \mu A_i^l}{\sum_{i=1}^m \prod_{i=1}^n \mu A_i^l} \quad (5)$$

where i is the route number, m represents the total number of fuzzy rules (chosen as 16 in our application), n shows the number membership functions used for the input (chosen as 4 in our application), A_i^l is the fuzzy value corresponding to each function above and μ is a measure of the CH stability level evaluated at the destination node. At first, the system normalizes each parameter from its initial domain (α, β) to the standard interval $(0.01, 1)$, where α and β are the minimum a maximum value respectively. For example, if the value of a certain parameter is ϕ , the normalized value is determined by [18].

$$N = \frac{\phi - \alpha}{\beta - \alpha} \quad (6)$$

The rules set for the inference transforms the resulting fuzzy sets into the space of the final fuzzy solution. In this approach we considered three possible values set as low, medium, and high are presented in this method. Thus, in Fig. 4, 5, 6 and 7 are shown the fuzzy variable rules for the remining energy, centrality, distance between CH and mobile sink, and the number of rounds the CH works as head. In these figures the membership value and input variable is between 0 to 1, and each criteria has a corresponding domain. Next stage, the fuzzifier transforms the input values in corresponding fuzzy sets as established based on the corresponding membership functions.

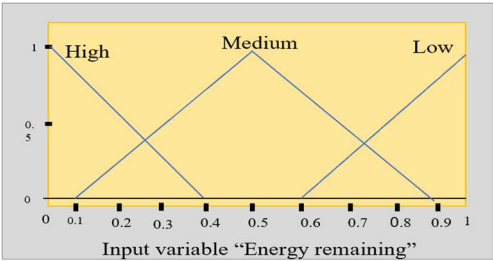


Fig. 4. Membership functions of system input variables in energy remaining

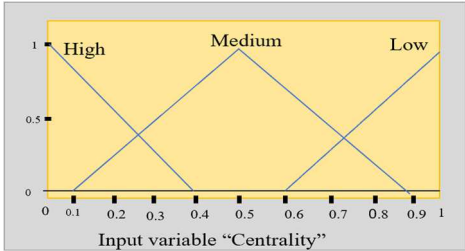


Fig. 5. Membership functions of system input variables in centrality

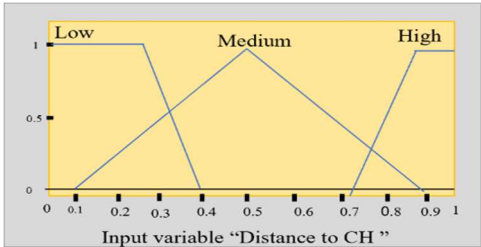


Fig. 6. Membership functions of system input distance between CH and mobile sink

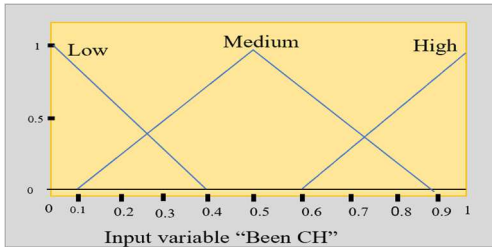


Fig. 7. Membership functions of system input the CH been as header

Table I. Fuzzy rule base

No.	Input				Output
	Distance to CH	Been CH	Energy remining	Centrality	
1	Low	Low	Low	Low	Low
2	Low	Low	Low	High	Low
3	Low	Low	High	Low	Low
4	Low	Low	High	High	Medium
5	Low	High	Low	Low	Low
6	Low	High	Low	High	Medium
7	Low	High	High	Low	Medium
8	Low	High	High	High	High

9	High	Low	Low	Low	Low
10	High	Low	Low	High	Medium
11	High	Low	High	Low	Medium
12	High	Low	High	High	High
13	High	High	Low	Low	Medium
14	High	High	Low	High	High
15	High	High	High	Low	High
16	High	High	High	High	High

IV. EXPERIMENTAL RESULTS

We used the MATLAB tool to simulate the proposed technique. A PC with 16 gigabytes of RAM and an Intel Core TM i7 3.3 GHz processor was used to run the simulations. The proposed approach was tested on a variety of values for the resources and requests in order to evaluate it. The simulation results show that, when compared to ad hoc on-demand distance vector (AODV) [19] method and with a fuzzy-based on-demand routing protocol (FBORP) [20], the proposed method is able to decrease the latency, and increase the reliability.

A. Simulation parameters

This paper investigates the algorithm described above in a homogenous network. The simulations were repeated 900 rounds. The circular network has a radius of 500 meters. There are 300 nodes in the network, each with an initial energy of $E_{init}=50J$, and the packet size was chosen to be 1024 bytes [21]. The IoT network parameters in MATLAB are described in Table II.

Table II. Simulation parameters

Parameter name	Value
Network radius(R)	500 m
Node velocity	(min = 5 m/s, max = 15 m/s)
Total number of nodes	300
Location of BS (sinks)	(250, 250)
Data packet size	1024 byte
Initial node energy	50 J
E_{elec}	50 nJ/bit
E_{amp}	$0.0013 \text{ pJ/bit/ } m^4$
Distance threshold (d_0)	$\sqrt{\frac{E_{elec}}{E_{amp}}} \text{ m}$

B. Obtained results

Within this section are shown the simulation results by comparing the performances obtained by the proposed method with the ones obtained by other well-known similar algorithms from the existing literature, based on energy consumption, delay, and reliability, as follows.

With respect to the delay, in figure 8 are shown the results obtained with respect to delay versus number of rounds. According to Figure 8, the FBORP achieves a higher delay with 16% then the proposed method, while the AODV obtains a higher delay with 11%. At the beginning, the result of the proposed method are close to AODV, but from round 400 to 600 proposed methods achieves a better result. The main

reasons for this high delay in AODV and FBORP are the clustering methods and algorithms used for choosing CH nodes. In these two techniques, the CH location is not considered, thus data transmission from CH to mobile sink has an increased delay. Given that in our proposed method one of the important criteria in choosing the CH is its location, it is expected that it achieves superior performance over other these criteria.

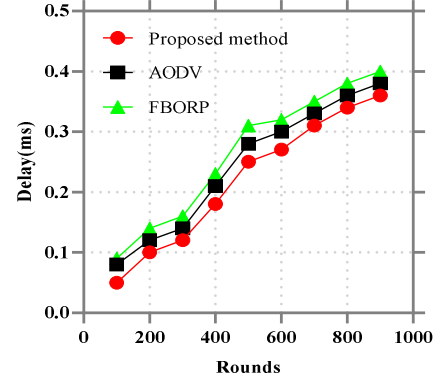


Fig. 8. Delay in 900 rounds

With respect to the total network energy consumption, the results are presented in figure 9 versus number of rounds. It can be observed that the maximum energy consumption is obtained by proposed method, followed closely by the AODV, while the smallest energy consumption is attained by FBORP. We can also observe that, at the beginning, the proposed method results are close to the ones achieved by AODV, but from round 700 the proposed method start achieving better results. The total AODV energy finishes after 830 rounds, and the proposed method finishes after 860 rounds while the FBORP energy finishes after 800 rounds.

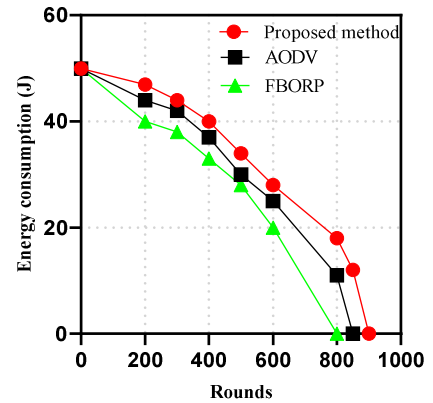


Fig. 9. Energy consumption in 900 rounds

Reliability is a feature of a system that may be used to estimate the likelihood of accomplishing tasks under established circumstances over a certain amount of time. Figure 10 illustrates the reliability dependance versus the number of rounds, and it can be easily seen that the suggested technique obtains better results than the other two methods, obtaining a higher reliability after all the 1000 rounds. Nonetheless, the

FBORP and AODV algorithms are very close to one another with respect to the reliability performances.

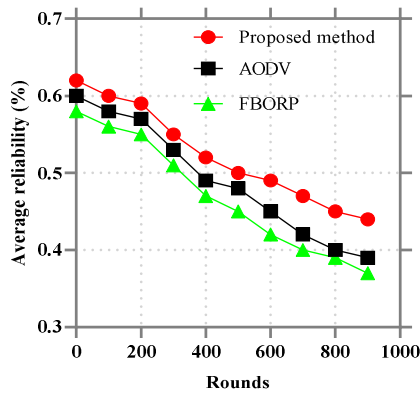


Fig. 10. Average reliability in 900 rounds

V. CONCLUSION AND FUTURE WORK

Improving URLLC in IoT an important issue, since it significantly impacts the network's running time and QoS. Important studies in the IoT field are related to clustered routing protocol and selection of the CH. In this article, we used the a combination of fuzzy logic algorithm in order to suitable select the CH. A node selection method was proposed, based on the remaining energy, centrality, distance to CH, and number of rounds in which one particular node have been head. In addition, the near-optimal intermediate node is employed for multi-hop routing utilizing the aforementioned technique. The suggested fuzzy logic approach was compared to AODV and FBOP in order to assess it. The proposed method proved its superiority with respect to the mentioned clustering strategies, shown by the simulated results. Other factors and techniques, such as Machine learning and the Artificial Fish-Swarm Algorithm (AFSA), may be utilized in the future to improve network performance. For the large scalability can use more than one mobile sink, so parallel computing can be future work for this area. Also, a priority of data gathering from the CH can be future work for research in this field.

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